

Particle motion in electric fields

X21 (2021) — English translation

Part A. Motion in a non-uniform field of constant direction (3.9 points)

A particle with charge q and mass m moves only under the influence of the electric field $E(y)$, parallel to the axis y along the trajectory $y = b\sqrt{1 - \frac{x^2}{a^2}}$ (the values a and b are also known $a, b > 0$). At the initial moment of time, the particle speed is v_0 and is directed perpendicular to the electric field.

A1	Find the dependence of the projection of the electric field $E_y(y)$ on the coordinate y , where $y \in (0, b]$. Express your answer using v_0, a, b, q, m .	2.40
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A2	From the same initial point in the same electric field, the same particle is released without an initial velocity. Determine the speed v_1 when the particle passes the distance $s = b/41$. Express your answer using v_0, a, b .	1.20
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A3	Find the travel time τ for the previous point. Express your answer in terms of the quantities given in A1.	0.30
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Part B. Charge Distribution (2.1 points)

B1	For convenience, we will continue to use this notation. Let $E_y(b) = E_0$. Write down how $E_y(y)$ from A1 is expressed in terms of E_0, b, y .	0.10
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Let now the electric field be arranged like this: $E = 0$ for $y > 2b$; $y < \frac{b}{100}$. For $y \in (b/100, 2b)$ the functional dependence found in point A1 is satisfied. Recall that the electric field is everywhere directed parallel to the Oy axis. This electric field is created by the following system of charges: Charged plane $y = y_1 = b/100$ with surface charge density σ_1 ; Charged plane $y = y_2 = 2b$ with surface charge density σ_2 ; A volume charge with a density of $\rho(y)$, distributed between the above planes.

B2	Find $\sigma_1, \sigma_2, \rho(y)$. Express your answer in terms of E_0, b, y and fundamental constants.	2.00
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Part C. Equipotential surfaces of a charged segment (4 points)

A rod of length $2L$ is charged uniformly along its length with a linear charge density ρ . Let us introduce a coordinate system YOX with the origin at the center of the rod, the axis OX is directed along the rod, and the axis OY is perpendicular to it.

C1 Find points lying on lines $x = 0$ and $y = 0$ such that the potential at these points is equal to φ_0 . **0.60**

C2 Prove that the curve $y(x)$ such that the potential at any point is equal to φ_0 is an ellipse. Find the equation of the given ellipse in coordinates $y(x)$, as well as its semi-major and minor axes a and b , respectively. **2.00**

Along a smooth spoke shaped like an ellipse, found at point C2, a charged particle of mass m and charge q is launched from point $x = 0$ at speed v_0 . The spoke potential is φ_0 .

C3 Find the radius of curvature R of the given ellipse at the point with coordinate x . Express your answer using x , a and b . Note: If the function $y = f(x)$ is given, then its radius of curvature at a given point can be calculated using the formula **0.40**

$$R = \frac{\left(1 + y'^2\right)^{\frac{3}{2}}}{|y''|}.$$

C4 Find the force of interaction between the particle and the spoke at its points of intersection with the straight line $x = 0$. Express your answer in terms of $m, q, v_0, L, \rho, \varphi$ and physical constants. **0.50**

C5 Find the force of interaction between the particle and the spoke at its points of intersection with the straight line $y = 0$. Express your answer in terms of $m, q, v_0, L, \rho, \varphi$ and physical constants. **0.50**

Source: <https://pho.rs/p/103>